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INFORMATION PROCESSING DEVICE TRANSMITTING FINALIZED PRINT DATA TO PRINTER BASED ON ORIGINAL PRINT DATA RECEIVED FROM STANDARD PRINTING SOFTWARE AS RESPONSE TO REQUEST

Abstract

An information processing device includes a storage and a processor. The storage stores an information processing program supporting a printer. The processor executing the program receives specified print data from a standard printing software integrated in an operating system. The standard printing software performs a modification process modifying the original print data to size-reduced print data representing a size-reduced image when a size of an original image represented by original print data exceeds a designated sheet size of a printing sheet for printing. The specified print data is the original print data or the size-reduced print data. In response to reception of the specified print data, the processor executing the program transmits, to the standard printing software, a request for the original print data, receive the original print data from the standard printing software, and transmit, to the printer, finalized print data based on the original print data received.

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Background/Summary

REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority from Japanese Patent Application No. 2024-033166 filed on Mar. 5, 2024. The entire content of the priority application is incorporated herein by reference.

BACKGROUND ART

[0002] A technology has been disclosed for switching the sheet size setting that has priority between the setting provided in the printer driver and the setting established on the printer body itself when the size of the sheet being printed is indefinite.

SUMMARY

[0003] In recent years, operating systems possessing a function known as a driverless printer function (OS-standard printing software) are becoming more commonplace for printing according to a predetermined printing standard without using a printer driver provided by the vendor of the printer. In such cases, a user can perform printing on the printer without having to install a printer driver.

[0004] However, when using the driverless function of the OS-standard printing software described above, the conventional technology can only handle sheet size settings generically and does not support printer-specific settings. For example, one function of the OS-standard printing software may be to reduce the image size of print data automatically when the length of the image specified by the print data exceeds the length set in the sheet size setting for the sheet to be printed.

[0005] Now, consider a conceivable case in which the user intends to print on printing sheet of an indefinite length, such as continuous length label sheet. In this conceivable case, assuming the OS-standard printing software does not support a size setting of indefinite length and must always require that some fixed size setting be specified, the software will view the length specified by some print data as exceeding the length of the size setting and will reduce the size of the image to be printed. In such cases, the user cannot obtain the expected printing results.

[0006] In view of the foregoing, it is an object of the present disclosure to provide an information processing program and an information processing device that can perform suitable printing on printing sheet of an indefinite length while supporting driverless OS-standard printing software.

[0007] In view of the foregoing, it is an object of the present disclosure to provide an image forming apparatus a non-transitory computer readable storage medium storing an information processing program executed by a computer included in an information processing device. The information processing device has an operating system installed thereon. The operating system has standard printing software integrated therein. The standard printing software is configured to perform a modification process when a size of an original image represented by original print data exceeds a designated sheet size of a printing sheet for use in printing the original print data. The modification process includes modifying the original print data to size-reduced print data representing a size-reduced image derived by reducing the size of the original image. The information processing program supports a printer communicable with the information processing device. The information processing program, when executed by the computer, causes the information processing device to perform: receiving specified print data from the standard printing

software, the specified print data being one of the original print data and the size-reduced print data; in response to reception of the specified print data, transmitting, to the standard printing software, a request for the original print data; receiving the original print data from the standard printing software as a response to the request; and transmitting, to the printer, finalized print data based on the original print data received as the response to the request.

[0008] In order to attain the above and other objects, the present disclosure provides an information processing device. The information processing device includes a communication interface communicable with a printer, a storage, and a processor. an information processing program supporting the printer and stored in the storage, when executed by the processor, causes the processor to perform: receiving specified print data from a standard printing software, the standard printing software integrated in an operating system installed on the information processing device, wherein the standard printing software is configured to perform a modification process when a size of an original image represented by original print data exceeds a designated sheet size of a printing sheet for use in printing the original print data, wherein the modification process includes modifying the original print data to size-reduced print data representing a size-reduced image derived by reducing the size of the original image, the specified print data being one of the original print data and the size-reduced print data; in response to reception of the specified print data, transmitting, to the standard printing software, a request for the original print data; receiving the original print data from the standard printing software as a response to the request; and transmitting, to the printer through the communication interface, finalized print data based on the original print data received as the response to the request.

[0009] In the above structures, suitable printing on indefinite length printing sheet can be performed using the OS-standard printing software without using a printer driver.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0010] FIG. 1 is a block diagram illustrating a functional configuration of a printing system including a printer and a terminal device.

[0011] FIG. 2 is a flowchart illustrating a control procedure according to a comparative example performed by an editing application, CUPS, printer application, and printer work in conjunction with each other.

[0012] FIG. 3 is a flowchart illustrating a control procedure according to an embodiment performed by an editing application, CUPS, printer application, and printer work in conjunction with each other.

[0013] FIG. 4 is a flowchart illustrating a control procedure according to a variation (1) performed by an editing application, CUPS, printer application, and printer work in conjunction with each other.

[0014] FIG. 5 is a flowchart illustrating a control procedure according to a variation (2) performed by an editing application, CUPS, printer application, and printer work in conjunction with each other.

[0015] FIG. 6 is a flowchart illustrating a control procedure according to a variation (3) performed by an editing application, CUPS, printer application, and printer work in conjunction with each other.

[0016] FIG. 7 is a block diagram illustrating a hardware configuration of the printing system.

DESCRIPTION

Embodiment

[0017] An embodiment of the present disclosure will be described while referring to the accompanying drawings.

Overview of a Printing System

[0018] FIG. 1 shows a sample configuration of a printing system 1 according to the present embodiment. In FIG. 1, the printing system 1 has a terminal device 100, and a printer 200. The terminal device 100 and printer 200 are connected and can communicate with each other via an internet, wired LAN, or wireless LAN, for example. The terminal device 100 is an example of the information processing device. The printer 200 is an example of the printer or printing device.

Functional Configuration of the Terminal Device

[0019] The terminal device 100 sends print data to the printer 200 to execute printing. The terminal device 100 is a desktop PC, a laptop, or a tablet computer, for example, but may also be a smartphone or other portable terminal. While described later in detail (see FIG. 6), the terminal device 100 has a processor 110, and a storage 115 storing various printing-related programs. With the processor 110 configured to execute the various printing-related programs stored in the storage 115, the terminal device 100 has the functional configuration shown in FIG. 1.

[0020] As shown in FIG. 1, the terminal device 100 has an operating system (OS) 101, an editing application 102, a printer application 170, and a browser 104. The OS 101 has a function known as a driverless printing function for printing according to a predetermined printing standard without using a printer driver provided by the vendor of the printer.

[0021] In this example, the OS 101 possesses the Common Unix Printing System (CUPS) 160, which is an OS modular printing system. The OS 101 is an example of an operating system and the CUPS 160 is an example of the OS-standard printing software or the standard printing software integrated in the OS and an example of the software built into the OS (the OS built-in software).

CUPS

[0022] The CUPS 160 includes an HTTP server 161, a CUPS PnP manager 162, an IPP client 163, and a print data scheduler and spooler 164.

[0023] The HTTP server 161 is server software that runs on the CUPS 160. The HTTP server 161 uses the Hypertext Transfer Protocol (HTTP) to acquire and provide information and functions through the browser 104. The CUPS PnP manager 162 has a plug and play (PnP) function for creating printer queues by detecting printers (including virtual printing devices) using a protocol such as multicast Domain Name System Service Discovery (mDNS-SD) and registering them in the CUPS 160.

[0024] The configuration of the CUPS PnP manager 162 is modified according to a script executed when the printer application 170 is installed in order that addresses searched using PnP are restricted to the internal network of the terminal device 100 (normally IP address 127.0.0.1).

Accordingly, the CUPS 160 cannot directly recognize the printer 200 but can recognize the printer application 170 as a printer. The printer application 170 is not an actual printer but is known as a virtual printer that behaves as a printer to the CUPS 160. The printer application 170 acts as an intermediary between the CUPS 160 and printer 200 and facilitates printing using the CUPS 160 and printer 200. The printer application 170 is an example of the virtual printer or virtual printing device.

[0025] The IPP client 163 exchanges print data and attributes with an IPP/mDNS server 173 described later using the Internet Printing Protocol (IPP).

[0026] The print data scheduler and spooler 164 manages print jobs received from the editing application 102 and the like. The print data scheduler and spooler 164 manages a plurality of print jobs generated simultaneously by temporarily queuing and sequentially processing the print jobs before transferring them to the IPP client 163.

Printer Application

[0027] The printer application 170 can be registered in the CUPS 160 as a virtual printer supporting the printer 200 connected to the terminal device 100. The CUPS 160 treats the printer application 170 as a virtual printer to execute various processes related to printing. These processes include a capability information acquisition process and a print data transmission process, for example. The

printer application **170** includes a printer communication interface (IF) **171**, an attribute/mDNS converter **172**, an IPP/mDNS server **173**, and a print data converter **174**. The printer application **170** is an example of the information processing program.

[0028] The printer communication interface **171** communicates with the printer **200** using the IPP protocol. The printer communication interface **171** receives attributes from the printer **200** and transmits print data to the printer **200**.

[0029] The attribute/mDNS converter **172** acquires attributes representing capabilities of the printer **200** via the printer communication interface **171**. The attribute/mDNS converter **172** converts the attributes acquired from the printer **200** to attributes corresponding to printing functions that the CUPS **160** supports.

[0030] The attribute/mDNS converter **172** has a multicast Domain Name System (mDNS) function. The mDNS sends DNS queries within the local network as an Internet Protocol (IP) multicast to query all devices on the same network simultaneously. When receiving a query, devices on the network respond by sending their own IP address.

[0031] The IPP/mDNS server **173** outputs the attributes converted by the attribute/mDNS converter **172** to the CUPS **160** as part of the capability information acquisition process performed by the CUPS **160**. The CUPS **160** outputs to the IPP/mDNS server **173** print data corresponding to the attributes converted by the IPP client **163**. The CUPS **160** transmits to the IPP/mDNS server **173** the print data corresponding to the converted attributes in the print data transmission process. Then, the IPP/mDNS server **173** acquires from the CUPS **160** the print data corresponding to the converted attributes, and outputs this print data to the print data converter **174**. The IPP/mDNS server **173** has an mDNS function similar to the attribute/mDNS converter **172**.

[0032] The print data converter **174** acquires print data from the CUPS **160** via the IPP/mDNS server **173**. The print data converter **174** converts the print data acquired from the CUPS **160** to print data supported by the printing function of the printer **200**. The print data converter **174** outputs the converted print data to the printer communication interface **171**.

[0033] The printer communication interface **171** transmits print data outputted from the print data converter **174** to the printer **200**.

Printer

[0034] The printer **200** prints on printing sheets (or papers) based on print jobs received from the printer communication interface **171** of the terminal device **100**. The printer **200** uses sheet of an indefinite length as printing sheet (or printing paper). Here, the sheet of indefinite length is a sheet having a size having a definite length in one of horizontal and vertical directions. The printer **200** is a printing device that supports printing sheet of an indefinite length. For example, the printer **200** is a label printer that prints on continuous length sheet wound in a roll to create printed labels. However, the printer **200** may be a printing device other than a label printer that supports printing sheet of an indefinite length.

Features of the Embodiment

[0035] After editing print data with the editing application **102**, the user also sets the size of printing sheet (sheet size) and issues an instruction to execute a print. Hereinafter, the size of image or sheet has a vertical length (or a vertical dimension) and a horizontal length (or a horizontal dimension). Here, the vertical dimension of the image specified in the print data may be called the height while the horizontal dimension may be called the width. When the user is attempting to print on printing sheet of an indefinite length, such as continuous length label sheet (or paper), the user sets a sheet size corresponding to the indefinite length type of sheets (or papers). The sheet size of the indefinite length type of sheets has a definite dimension in one of the vertical and horizontal dimensions and an indefinite dimension in remaining one of the vertical and horizontal dimensions. However, the CUPS **160** may not support an indefinite length size setting and may require that a size setting of some type of definite length (fixed length) settings, such as prescribed sizes of die-cut labels, A3, A4, and letter, must be specified. When the user selects the size of indefinite length

type of sheet, the editing application **102** changes the user selected size to an acceptable size that the CUPS **160** can recognize, and transmits the acceptable size to the CUPS **160**. For example, the user selects a size having **26** mm height×indefinite width, the editing application **102** changes this size to an acceptable size of 26 mm height×100 mm width by replacing the indefinite width with a predetermined definite dimension (100 mm in this example). Thereafter, the editing application **102** sends to the acceptable size having 26 mm height×100 mm width to the CUPS **160**. In such cases, the size of the image specified in the print data may exceed the selected size of the printing sheet. Hereinafter, the size, length, and dimensions of the image specified in the print data is simply referred to as the size, length, and dimensions specified in the print data. When the size specified in the print data exceeds the size of the printing sheet, the CUPS **160** may perform a function to reduce the size of image by modifying the print data automatically. Reducing the size of image by modifying the print data may be referred to reducing the print data in dimensions. When print data is reduced in dimensions by such a function of the CUPS **160**, the user may not obtain the expected printing results because the actual sheet size has indefinite length and reducing the print data in dimensions is not necessary.

[0036] Hereinafter, the sheet size of indefinite length may be referred to as the indefinite sheet size or the indefinite size, and the sheet size that is not the indefinite sheet size and thus has definite dimensions in vertical and horizontal directions may be referred to the definite sheet size or the definite size.

[0037] Therefore, to account for a possibility that the actual sheet size for sheet to be printed is a size having indefinite length, the printing system **1** of the present embodiment performs printing based on the original print data outputted from the editing application **102** and saved in the CUPS **160** instead of the print data that the CUPS **160** has outputted intended for printing. The CUPS **160** in the present embodiment has specifications that do not support an indefinite length size setting and requires that some size setting of the definite sizes (fixed sizes) be specified. Hence, it is possible that the user intended to specify indefinite length, even when some other size setting has been made. For this reason, the printing system **1** of the present embodiment performs printing based on the original print data instead of the print data outputted from the CUPS **160** intended for printing, regardless of the sheet size setting specified by the user. As a result, the printing system **1** can perform suitable printing for indefinite length printing sheet and can create printed matter with content in line with the user's intentions and without any size reductions in dimensions not intended by the user. This process is described below in detail.

Control Procedure

[0038] First, a comparative example in which printing is not performed based on the original print data will be described according to the flowchart in FIG. **2** as a comparison to the present embodiment. The editing application **102**, CUPS **160**, printer application **170**, and printer **200** work in conjunction with each other to execute the control procedure in this comparative example.

[0039] First, in S5 of FIG. **2** the editing application **102** accepts user operations for editing print data and selecting a size for the printing sheet. For the size selection, the editing application **102** offers, as options, a plurality of sizes including 26×51 mm, 26×76 mm, and 50×102 mm, and the user selects one of these sizes. The editing application **102** also offers one or more options each indicating a sheet size having indefinite length. The options may be provided in various formats including: actual size dimensions “26 mm height×51 mm width” or “26 mm × indefinite width”; identifiers or names such as “A4” or “continuous label sheet”; and graphical representations or images depicting the sheet sizes. For indicating the sheet size of definite length x indefinite length, such as “26 mm×indefinite width”, the only one definite dimension of two dimensions, such as “26 mm” may be presented as the option. In other words, the option specifying only one definite dimension (such as “26 mm”) may be identifier indicating that the sheet size is of indefinite type.

[0040] As described above, when the user selects an option indicating a sheet size having indefinite length, such as 26 mm height×indefinite width, the editing application **102** changes this sheet size

to an available sheet size, such as 26 mm height×100 mm width, by replacing an indefinite dimension of the user selected sheet size with the predetermined definite dimension such as 100 mm.

[0041] In **S10** the editing application **102** determines whether the user has issued an instruction to execute a print. The editing application **102** repeats the determination in **S10** while an instruction to print has not been received (**S10**: NO). When the user has issued an instruction to execute a print (**S10**: YES), the process advances to **S15**.

[0042] In **S15** the editing application **102** transmits a first print job to the CUPS **160**. The first print job includes the print data edited in **S5**, sheet size information, a job ID, and the like. The sheet size information included in the first print job indicates the user selected sheet size selected in **S5** when the user selected sheet size indicates the definite size of sheet that has definite dimensions in the vertical and horizontal dimension. However, the sheet size information indicates the available sheet size that the editing application **102** generates based on the user selected sheet size when the user selected sheet size indicates the sheet size of indefinite length.

[0043] In **S20** the CUPS **160** receives the first print job generated and sent by the editing application **102**. The CUPS **160** saves the first print job in a suitable storage area of the storage **115** (see FIG. 6).

[0044] In **S25** the CUPS **160** determines whether the size of the image specified in the print data contained in the first print job received in **S20** exceeds the sheet size specified by the sheet size information in the received first print job. When each of vertical and horizontal dimensions specified in the print data is within the corresponding dimension of the sheet size (**S25**: NO), the CUPS **160** advances to **S35** described later. However, when at least one of the vertical and horizontal dimensions specified in the print data are greater than the corresponding dimension of the sheet size (**S25**: YES), the CUPS **160** advances to **S30** described below.

[0045] In **S30** the CUPS **160** reduces the size of the image of the print data based on the first print job received in **S20** to a size that fits within the specified size for the printing sheet. Specifically, the CUPS **160** modifies the print data so that both vertical and horizontal dimensions of the image represented by the print data can fit within the corresponding dimensions of the printing sheet by reducing the dimensions of the image while maintaining a constant aspect ratio of the image.

[0046] When the CUPS **160** has reached a NO determination in **S25**, in **S35** the CUPS **160** transmits a second print job to the printer application **170** based on the print data received in **S20**, i.e., print data that has not been reduced in dimensions. The second print job in this case (**S25**: NO) includes the print data received in **S20**, and the sheet size information and job ID included in the first print job. When the CUPS **160** reached a YES determination in **S25**, the CUPS **160** transmits the second print job to the printer application **170** based on the modified print data whose image has the reduced size as a result of the process of **S30**. That is, the second print job in this case (**S25**: YES) includes the modified print data as a result of **S30**, and the sheet size information and job ID included in the first print job.

[0047] As a specific example, when 50 mm width×102 mm height has been selected as the size of the printing sheet and 50 mm width×100 mm height is the image size specified in the print data, the CUPS **160** transmits a second print job to the printer application **170** based on print data of the original size. On the other hand, when the size specified in the print data is 50 mm width×200 mm height with the same size (50 mm width×102 mm height) of the printing sheet selected, the CUPS **160** reduces the size of the image in the print data from a 200 mm height to a 102 mm height while maintaining the aspect ratio and transmits a second print job to the printer application **170** based on the modified print data that has been reduced in dimensions. In this case, the width specified in the print data is also reduced to about one half. As another example, when the size specified in the print data is 100 mm width×100 mm height with the same size (50 mm width×102 mm height) of the printing sheet selected, the CUPS **160** reduces the size of the image in the print data from a 100 mm width to a 50 mm width while maintaining the aspect ratio and transmits a second print job to

the printer application **170** based on the print data that has been reduced in dimensions. In this case, the height of the print data is also reduced to one half.

[0048] In **S40** the printer application **170** receives the second print job from the CUPS **160**.

[0049] In **S45** the printer application **170** converts the print data included in the second print job received in **S40** to print data supported by the printing function of the printer **200**. For example, the print data converter **174** of the printer application **170** performs rasterization to convert the print data from vector data to raster data.

[0050] In **S50** the printer application **170** transmits a finalized print job to the printer **200** based on the print data converted in **S45**. The finalized print job includes the print data converted in **S45**, the sheet size information, the job ID, and the like.

[0051] In **S55** the printer **200** receives the finalized print job generated by the printer application **170**.

[0052] In **S60** the printer **200** prints the print data based on the finalized print job received in **S55**. This completes the procedure shown in the flowchart of FIG. 2.

[0053] In the comparative example described above, the CUPS **160** reduces the size of the image of the print data when the size specified in the print data is greater than the selected size of the printing sheet. Since the printer application **170** controls the printer **200** to print based on this print data that has been reduced in dimensions, the user may not obtain desired printing results.

[0054] Next, a control procedure according to the present embodiment will be described with reference to the flowchart in FIG. 3. The editing application **102**, CUPS **160**, printer application **170**, and printer **200** work in conjunction with each other to execute this control procedure.

[0055] In **S105** of FIG. 3, first the editing application **102** accepts user operations to edit print data and to select the size of the sheet to be printed. **S105** is performed similarly to **S5**. Specifically, when the user selects an option indicating a sheet size having indefinite length, such as 26 mm height×indefinite width, the editing application **102** changes this sheet size to an available sheet size, such as 26 mm height×100 mm width, by replacing an indefinite dimension of the user selected sheet size with the predetermined definite dimension such as 100 mm.

[0056] In **S110** the editing application **102** determines whether the user issued an instruction to execute a print. The editing application **102** repeats the determination in **S110** while an instruction to print has not been received (**S110**: NO). When an instruction to execute a print is received (**S110**: YES), the process advances to **S115**.

[0057] In **S115** the editing application **102** transmits a first print job to the CUPS **160**. The first print job includes the print data edited in **S105**, and sheet size information and the job ID. Similarly to **S15**, the sheet size information included in the first print job indicates the user selected sheet size is the sheet size selected in **S5** when the user selected sheet size indicates the definite size of sheet that has definite dimensions in the vertical and horizontal dimension. However, the sheet size information indicates the available sheet size that the editing application **102** generates based on the user selected sheet size when the user selected sheet size indicates the sheet size of indefinite length.

[0058] In **S120** the CUPS **160** receives the first print job generated by the editing application **102**. The CUPS **160** extracts the print data from the first print job received in **S120** and saves this print data in a suitable storage area of the storage **115**. In this case, the CUPS **160** may save the print data in association with the job ID in the first print job in the storage area. In this embodiment, the print data extracted and saved in a storage area in **S120** will be called the “original print data” for convenience. In **S120** the CUPS **160** may save the first print job exactly as received in the storage **115**.

[0059] In **S125** the CUPS **160** determines whether the size of the image represented by the print data generated above in **S120** exceeds the sheet size specified by the sheet size information in the received first print job. When each of the height and width specified in the print data is within a corresponding one of the height and width of the sheet size (**S125**: NO), the CUPS **160** advances to

S135 described later. On the other hand, when at least one of the length and width specified in the print data is greater than a corresponding one of the length and width of the sheet size (**S125: YES**), the CUPS **160** advances to **S130** described next.

[0060] In **S130** the CUPS **160** modifies the print data generated in **S120** to a size that fits within the selected size of the printing sheet. At this time, the CUPS **160** modifies the print data so that each of the length and width dimensions can fit within a corresponding one of the length and width of the printing sheet while maintaining a constant aspect ratio. The print data modified in **S130** is an example of the size-reduced print data. The size or dimensions of the image represented in the print data may be determined by the number of pixels in each of the vertical and horizontal directions and resolution specifying the number of pixels that appears in a unit length such as dot per inch (DPI). The information on the resolution may be included in the print data. Alternatively, the information on the resolution may be included in a print job including the print data. In **S130**, the CUPS **160** may modify print data by reducing the numbers of pixel in each of the vertical and horizontal directions and/or increasing the resolution in each of the vertical and horizontal directions.

[0061] When the CUPS **160** reached a NO determination in **S125**, in **S135** the CUPS **160** transmits a second print job to the printer application **170** based on the print data generated in **S120**, i.e., print data that has not been reduced in dimensions. The second print job in this case (**S125: NO**) includes the print data extracted in **S120**, and the sheet size information and job ID included in the first print job. On the other hand, when the CUPS **160** reached a YES determination in **S125**, in **S135** the CUPS **160** transmits a second print job to the printer application **170** based on the print data reduced in dimensions in **S130**. The second print job in this case (**S125: YES**) includes the modified print data modified in **S130** to represent the image having the reduced size, and sheet size information and job ID included in the first print job.

[0062] In **S140** the printer application **170** receives the second print job generated by the CUPS **160**. The print data included in the second print job is an example of the specified print data.

[0063] In **S145** the printer application **170** extracts the job ID contained in the second print job received in **S140**.

[0064] In **S150** the printer application **170** sends a transmission request to the CUPS **160** requesting the CUPS **160** to transmit the original print data based on the job ID acquired in **S145**. The job ID serves to identify the original print data, which is the subject of the transmission request. The process of step **S150** is executed upon reception of the second print job in **S140**. In other words, the printer application **170** sends a transmission request for the original print data regardless of what size of printing sheet has been designated by the sheet size information in the second print job. The printer application **170** is designed in advance to have access privileges for requesting the CUPS **160** to transmit original print data. The transmission request is an example of the request.

[0065] In **S155** the CUPS **160** receives the transmission request for the original print data from the printer application **170**.

[0066] In **S160** the CUPS **160** reads, from the storage **115**, the original print data identified by the job ID in the transmission request received from the printer application **170** and transmits a third print job including the read print data to the printer application **170**. Here, the third print job includes the original print data that has been saved in a storage area in **S120**, and the sheet size information and job ID included in the second print job and job ID the same as the job ID included in the transmission request.

[0067] In **S165** the printer application **170** receives the third print job containing the original print data from the CUPS **160** in response to the transmission request sent in **S150**. The receiving the third print job is an example of the receiving the original print data.

[0068] In **S170** the printer application **170** converts the original print data contained in the print job received in **S165** to print data supported by the printing function of the printer **200**. As in **S45** in FIG. 2, in **S170** the printer application **170** performs rasterization to convert the print data from

vector data to raster data, for example. In **S170** the printer **170** converts the original print data in such a manner that the size of the image in the converted print data is unchanged and thus is the same as the size of the image in the original print data. The process of **S165** is an example of the original data acquisition process.

[0069] In **S175** the printer application **170** transmits a finalized print job based on the original print data acquired in **S165** and converted in **S170** to the printer **200**. The finalized print job includes the original print data acquired in **S165** and converted in **S170**, and sheet size information and job ID. The finalized print job may not include the sheet size information. The original print data transmitted in **S170** is an example of the finalized print data, and the process of **S175** is an example of the transmitting the finalized print data.

[0070] In **S180** the printer **200** receives the finalized print job generated and sent by the printer application **170**.

[0071] In **S185** the printer **200** prints the print data based on the finalized print job received in **S180**. This completes the process in the flowchart of FIG. 3. The printer **200** may not reference the sheet size information included in the finalized print job, and perform the printing based on a sheet installed in the printer **200**.

[0072] The printer **200** may have a function to modify the print data included in the finalized print job so that the image represented in the print job is reduced in dimensions. For example, when the finalized print job includes the original print data and the size of the image specified in the original print data exceeds the size of the printing sheet, the printer **200** may modify the print data so that the reduced size of the image can fit the size of the printing sheet. In this reduction process, when the size of the sheet that the printer **200** uses has an indefinite dimension, the printer **200** considers this indefinite dimension. That is, when the size of the sheet to be used has an indefinite dimension in one of the horizontal and vertical direction and a definite dimension in a remaining one of the horizontal and vertical directions, the printer **200** determines whether the dimension specified in the print data in the remaining one of the horizontal and vertical directions exceeds the corresponding dimension of the sheet to be used, and performs the reduction process when and only when the dimension specified in the print data in the remaining one of the horizontal and vertical directions exceeds the corresponding size of the sheet to be used.

Effects of the Embodiment

[0073] In the embodiment described above, the printer application **170** executes the process of **S140** to acquire print data generated and outputted by the CUPS **160**. The CUPS **160** has a function for reducing the size of the image in the print data when the size of the image represented by the print data exceeds the size of the printing sheet. In **S140** the printer application **170** acquires either print data that has not undergone such reduction in dimensions or print data that has undergone a reduction in dimensions.

[0074] To account for a possibility that the sheet to be printed is of indefinite length, the printer application **170** executes the processes of **S150** and **S165** upon acquiring the print data in **S140**. In **S150** the printer application **170** outputs a request to the CUPS **160** requesting the transmission of the original print data. In **S165** the printer application **170** newly acquires the original print data outputted from the CUPS **160** in response to the transmission request. In **S175** the printer application **170** sends the finalized print job to the printer **200** based on the original print data acquired in **S165**.

[0075] Thus, when printing on printing sheet of indefinite length in the present embodiment, the printer application **170** does not use the print data obtained in **S140**, which may have been reduced in dimensions, but ultimately uses the original print data acquired in **S165** as the finalized print data. As a result, the terminal device **100** of this embodiment can perform suitable printing on indefinite length printing sheet to create printed matter whose content is in line with the user's intention and that has not been reduced in dimensions against the wishes of the user.

[0076] A particular feature of this embodiment is that the printer application **170** executes steps

S150 and **S165** no matter what selection has been made for the size of the printing sheet.

Ultimately, the printer application **170** always uses, the finalized print data, the original print data acquired in **S165** rather than the print data acquired in **S140**. Thus, the terminal device **100** according to this embodiment can reliably perform printing suitable for indefinite length printing sheet.

[0077] Another feature of the present embodiment is that the printer application **170** is registered in the CUPS **160** as a virtual printer. By registering the printer application **170** as a virtual printer, the CUPS **160** can output print data to the printer application **170**, and the printer application **170** can control the printer **200** to perform a print based on that print data.

[0078] Another feature of the present embodiment is that in **S175** the printer application **170** transmits print data to the printer **200**, which is a label printer that creates printed labels by printing on a continuous sheet of printing sheet wound into a roll. Since a label printer creates printed labels using continuous printing sheet from a roll, for example, indefinite length is selected as the size of the printing sheet. Accordingly, when creating printed labels in this embodiment, the printer application **170** does not use the print data obtained in **S140** but ultimately uses the original print data acquired in **S165**. As a result, the printing system **1** according to this embodiment can perform suitable printing on a label printer by creating printed labels having content in line with the user's expectations and without any reduction in dimensions not in accordance with the user's wishes.

[0079] While the invention has been described in conjunction with various example structures outlined above and illustrated in the figures, various alternatives, modifications, variations, improvements, and/or substantial equivalents, whether known or that may be presently unforeseen, may become apparent to those having at least ordinary skill in the art. Accordingly, the example embodiments of the disclosure, as set forth above, are intended to be illustrative of the invention, and not limiting the invention. Various changes may be made without departing from the spirit and scope of the disclosure. Therefore, the disclosure is intended to embrace all known or later developed alternatives, modifications, variations, improvements, and/or substantial equivalents. Some specific examples of potential alternatives, modifications, or variations in the described invention are provided below:

(1) Using Original Print Data Only in Place of Print Data That Has Been Reduced in Dimensions

[0080] In the above embodiment, the printer application **170** obtains the original print data and uses this data in place of the print data to execute a print, regardless of what sheet size has been designated in the acquired second print job, i.e., regardless of whether the print data has been reduced in dimensions by the CUPS **160**. In this case, the original print data is acquired and used in place of the print data even when the print data has not been reduced in dimensions by the CUPS **160** and can be used as is. In this variation, the printer application **170** may replace print data with the original print data so that printing is performed based on the original print data, only when the print data has been reduced in dimensions by the CUPS **160**. Steps in the control procedure of this variation will be described next with reference to the flowchart in FIG. 4, wherein steps similar to those of FIG. 3 described above are designated with the same step numbers to avoid duplicating description.

[0081] Steps **S105-S140** of FIG. 4 are identical to those in FIG. 3, and a description of these steps will not be repeated here.

[0082] In **S141** the printer application **170** converts the print data contained in the second print job received in **S140** to print data supported by the printing function of the printer **200**. The printer application **170** converts the print data in **S141** in the same manner as **S170** shown in FIG. 3.

[0083] In **S142** the printer application **170** acquires the sheet size from the sheet size information in the second print job received in **S140**. The process of **S142** is an example of the acquiring the designated sheet size.

[0084] Subsequent steps **S145-S165** are identical to those in FIG. 3, and a description of these steps will be simplified. That is, the printer application **170** sends a transmission request to the CUPS

160 in **S150** for the original print data, regardless of the size selection results obtained in **S142**, and receives the third print job containing the original print data in **S165**.

[0085] In **S166** the printer application **170** extracts the size of the image represented by the original print data included in the print job received in **S165**.

[0086] In **S167** the printer application **170** determines whether the size specified in the original print data exceeds the size of the printing sheet based on the size specified in the original print data extracted in **S166** and the size of the printing sheet acquired in **S142**. Since the determination in **S167** is based on data that is essentially identical to the data used in **S125**, the determination in **S167** produces the same result as in **S125**. When the printer application **170** determines that the size specified in the original print data exceeds the size of the printing sheet (**S167**: YES), the process advances to step **S170**. The process of **S167** is an example of the determining whether the size of the original image represented by the original print data exceeds the designated sheet size.

[0087] Steps **S170** and **S175** are identical to those in FIG. 3, and a description of these steps will be simplified. That is, when the printer application **170** determines in **S167** that the size of the original print data is larger than the size of the printing sheet, in **S175** the printer application **170** transmits the finalized print job to the printer **200** based on the original print data acquired in **S165**.

[0088] On the other hand, when the printer application **170** determines in **S167** that the size specified in the original print data is no greater than the size of the printing sheet (**S167**: NO), the process advances to **S176**.

[0089] In **S176** the printer application **170** transmits a finalized print job to the printer **200** based on the print data acquired in **S140** and converted in **S141**. In other words, when the printer application **170** reaches a NO determination in **S167**, the print data received in **S140** can be used since the print data has not reduced in dimensions by the CUPS **160**. The finalized print job transmitted to the printer **200** includes the print data that has been acquired in **S140** and converted in **S141**, and sheet size information and job ID included in the second print job. The print data transmitted in **S176** is an example of the finalized print data based on the original print data received. The process of **S176** is an example of the transmitting the finalized print data. The process of **S141** may not be performed between **S140** and **S142**, and may be performed after a NO determination is made in **S167** and before performing **S176**.

[0090] In **S180** the printer **200** receives the finalized print job transmitted from the printer application **170** in **S175** or step **S176**.

[0091] In **S185** the printer **200** performs a print of the print data based on the finalized print job received in **S180**. This completes the process in the flowchart of FIG. 4.

[0092] In this variation, the printer application **170** executes steps **S150** and **S165** no matter what sheet size is designated in the sheet size information in the second print job. Subsequently, the printer application **170** executes the determination in **S167** on the original print data obtained in **S165**. When the size of the image represented by the original print data exceeds the size of the printing sheet, the printer application **170** does not include the print data acquired in **S140** in the finalized print job but includes the original print data obtained in **S165** in the finalized print job.

[0093] The printing system **1** according to this variation can reliably execute printing suitable for indefinite length printing sheet.

(2) Acquiring the Original Print Data Only When Indefinite Length Has Been Selected by the User

[0094] Similarly to the above embodiment, in this variation, when the user selects an option indicating a sheet size having indefinite length, such as 26 mm height×indefinite width, the editing application **102** changes this sheet size to an available sheet size, such as 26 mm height×100 mm width, by replacing an indefinite dimension of the user selected sheet size with the predetermined definite dimension such as 100 mm. This predetermined definite dimension (100 mm for example) is transferred from the editing application **102** to the printer application **170** as part of the print job. In this variation, the printer application **170** recognizes the predetermined definite dimension as an indicator of the indefinite dimension. That is, when the printer application **170** receives the sheet

size information indicating a sheet size having the predetermined definite dimension, the printer application **170** recognizes that the actual sheet size that the user has selected has an indefinite dimension. In other words, the predetermined definite dimension signifies that the dimension of the sheet size that the user has selected is actually the indefinite dimension. Put another way, the predetermined definite dimension is a dimension associated with the indefinite dimension. One or more predetermined definite dimensions may be registered in the storage **115** so as to specify that these predetermined definite dimensions are indicators of the indefinite dimension. The editing application **102** may refer to these predetermined definite dimensions in the storage **115** when generating the available size. The printer application **170** may refer to these the predetermined definite dimensions in the storage **115** when determining whether one dimension of the sheet size included the sheet size information matches the predetermined definite dimension to signify the actually-selected sheet size has indefinite dimension.

[0095] In this variation, the printer application **170** is configured to acquire the original print data only when the sheet size in the received sheet size information has a predetermined definite dimension signifying that the actually selected sheet size has an infinite dimension.

[0096] Steps in a control procedure of this variation will be described with reference to the flowchart in FIG. **5**, wherein steps similar to those in FIG. **4** have been designated with the same step numbers to avoid duplicating description.

[0097] Steps **S120-142** are identical those in FIG. **4**, and a description of these steps has been omitted.

[0098] In **S143** the printer application **170** determines whether that the sheet size selected by the user through the editing application **102** has an indefinite length. Specifically, the printer application **170** determines whether the sheet size acquired in **S142** has the predetermined definite dimension signifying indefinite length. In other words, in **S143** the printer application **170** determines whether one of the dimensions in the acquired sheet size from the sheet size information is an indicator indicating that the actually-selected sheet size has an indefinite dimension. When the sheet size indicated in the sheet size information in the second print job received in **S140** has the predetermined definite dimension, the sheet size indicated in the sheet size information included in the second print job signifies that the sheet size actually selected by the user has the indefinite length, and thus the printer application **170** determines that the sheet size selected by the user through the editing application **102** has an indefinite length. When the sheet size indicated in the second print job has no predetermined definite dimension, the printer application **170** determines that the sheet size selected by the user through the editing application **102** has no indefinite length. When the printer application **170** determines that the sheet size indicated in the second print job acquired in **S142** has the predetermined definite dimension associated with indefinite length (**S143: YES**), the process advances to step **S145**. When the printer application **170** determines that the sheet size has no predetermined definite dimension (**S143: NO**), the process advances to step **S176**. The process of **S143** is an example of the determining whether the designated sheet size is associated with a specific sheet size having an indefinite length.

[0099] The subsequent steps **S145-S165** and **S170-S185** are identical to those in FIG. **4**, and a description of these steps will be simplified. Unlike in the process of FIG. **4**, the printer application **170** advances directly to step **S170** following step **S165**. That is, the printer application **170** sends a transmission request to the CUPS **160** for the original print data in **S150** and transmits a finalized print job to the printer **200** in **S175** based on the original print data obtained in **S165**.

[0100] However, when the printer application **170** determined in **S143** that the sheet size indicated in the received sheet size information has no indefinite dimension and thus the user-selected sheet size has no indefinite length (**S143: NO**), in **S176** the printer application **170** transmits a finalized print job to the printer **200** based on the print data that has been obtained in **S140** and replaced in **S141** (the print data included in the second job). The subsequent steps **S180-S185** are identical to

those in FIG. 4.

[0101] In this variation, the printer application **170** executes steps **S150** and **S165** when the sheet size having indefinite length has been selected by the user as the size of the printing sheet. That is, the printer application **170** acquires the original print data only when the sheet size having indefinite length has been selected by the user as the size of the printing sheet and ultimately replaces the print data acquired in **S140** with the original print data acquired in **S165** to be used as the print data included in the finalized print job.

[0102] The printing system **1** according to this variation can reliably execute printing suitable for indefinite length printing sheet. Further, when the sheet size having indefinite length has not been selected by the user as the size of the printing sheet, the printing system **1** can ultimately use the print data acquired in **S140** and converted in **S141**. This variation can speed up the printing process by not unnecessarily performing steps **S145-S175** to replace the print data with the original print data.

(3) Acquiring the Original Print Data Only When The Print Job Includes An Indicator Indicating That Indefinite Length Has Been Selected

[0103] This variation (3) is based on the variation (2) shown in FIG. 5. Steps in the control procedure of this variation will be described next with reference to the flowchart in FIG. 6, wherein steps similar to those of FIG. 5 described above are designated with the same step numbers to avoid duplicating description.

[0104] In the variation (2) the printer application **170** makes the determination **S143** by using the predetermined definite length in the sheet size information as an indicator indicating that the sheet size actually selected by the user has the indefinite dimension. However, in this variation (3), instead of using the sheet size information, the printer application **170** makes a determination in **S243** (FIG. 5), which is similar to **S143**, by using additional information on an indicator indicating whether the sheet size actually selected by the user has the indefinite dimension or not. The indicator may be an identifier identifying a sheet size, such as a sheet-size name, for example. The storage **115** may store additional information for distinguishing whether each identifier indicates an indefinite sheet size or a definite sheet size. The indicator may be a value of a flag indicating whether a sheet size is a definite sheet size or an indefinite sheet size.

[0105] Specifically, after **S110**, in **S215** the editing application **102** transmits a first print job to the CUPS **160**. The first print job includes an indicator as well as the print data, the sheet size information, and the job ID. As described above, the indicator is information indicating whether the sheet size actually selected by the user has the indefinite dimension or not. The indicator may be information on an identifier indicating the sheet size actually selected by the user in **S105**. The indicator may be a name of the sheet size. For example, when the user-selected sheet size is a definite size having no indefinite dimension, the sheet-size name may be a character string directly (explicitly) indicating the definite size, such as, “definite size”, or a character string indicating a size or type of the sheet of the definite size such as “die-cut label”, “A3”, “A4” and “letter” so as indirectly (implicitly) indicate that the sheet size is the definite size having no indefinite dimension. When the user-selected sheet size is an indefinite size having an indefinite dimension, the sheet-size name may be a character string directly (explicitly) indicating the indefinite size, such as, “indefinite size” and “continuous length”, or a character string indirectly (implicitly) indicating a size or type of the sheet of the indefinite size, such as “label sheet”. The indicator may be a predetermined specific character or a predetermined specific string character that enables the printer application **170** to identify whether the sheet size that the user has selected is an indefinite sheet size or a definite sheet size. The indicator may be a value or a flag that enables the printer application **170** to at least distinguish whether a sheet size is an indefinite sheet size or a definite sheet size.

[0106] In **S120** the CUPS **160** receives the first print job transmitted in **S215** from the editing application **102**. The CUPS **160** extracts the print data from the first print job received in **S120** and

saves this print data in a suitable storage area of the storage **115**. Steps **S125** and **S130** are identical those shown in FIG. 5.

[0107] After **S130**, in **S235** the CUPS **160** transmits a second print job to the printer application **170**. The second print job includes the indicator, such as the sheet-size name, included in the first print job received in **S120**, as well as the print data, the sheet size information, and the job ID. Here, the print data included in the second print job is the print data extracted in **S120** when a NO determination is made in **S125**. However, when a YES determination is made in **S125**, the print data included in the second print job is the print data that has been reduced in dimensions in **S130**. [0108] In **S140** the printer application **170** receives the second print job transmitted in **S235**. Steps **S141** and **S142** are identical with those in FIG. 5.

[0109] After **S142**, in **S242** the printer application **170** extracts the indicator from the received second print job.

[0110] After **S242**, in **S243** the printer application **170** determines whether or not the indicator extracted in **S242** indicates that the user selected sheet size is the indefinite sheet size having an indefinite dimension. For example, when the indicator is the sheet-size name and the sheet-size name includes a character string indicating the definite sheet size, such as “die-cut label”, “A3”, “A4” or “letter”, the printer application **170** reaches a NO determination in **S245**. When a NO determination is made in **S243**, the process advances to **S176**. On the other hand, when the sheet-size name includes a character string indicating the indefinite size such as “indefinite size”, for example, the printer application **170** reaches a YES determination in **S243**. When a YES determination is made in **S243**, the process advances to **S145**. Steps **S145-S176**, and **S180-S185** are the same as those in FIG. 5.

[0111] In this variation, the printer application **170** executes steps **S150** and **S165** when the sheet size having indefinite length has been selected by the user as the size of the printing sheet. That is, the printer application **170** acquires the original print data only when the indicator indicates the sheet size having indefinite length, and replaces the print data acquired in **S140** with the original print data acquired in **S165** to be used as the print data included in the finalized print job.

[0112] The printing system **1** according to this variation can reliably execute printing suitable for indefinite length printing sheet. Further, when the sheet size having indefinite length has not been selected by the user as the size of the printing sheet, the printing system **1** can ultimately use the print data acquired in **S140** and converted in **S141** for the finalized print job. This variation can speed up the printing process by not unnecessarily performing steps **S145-S175** to replace the print data with the original print data.

(4) Other Variations

[0113] The above embodiment describes a case in which the printer application **170** is registered in the CUPS **160** as a virtual printer corresponding to the printer **200** connected to the terminal device **100**, but the present disclosure is not limited to this configuration. The printer application need not be registered in the CUPS **160** as a virtual printer but may simply serve to assist or relay information for printing processes performed between the CUPS **160** and printer **200**, for example. The information processing program of the present disclosure may be applied to such a printer application.

[0114] The above embodiment also describes a case in which the OS **101** possesses the CUPS **160** as an example of OS-standard printing software, but the present disclosure is not limited to this configuration. The OS **101** may possess OS-standard printing software other than CUPS, provided that the operating system includes a driverless printing function that uses the IPP protocol.

[0115] The flowcharts shown in FIGS. 2 through 6 in the above description do not limit the present invention to the steps indicated therein. Steps may be added or deleted, or their order may be rearranged, without departing from the scope and technical ideas of the invention.

[0116] In addition to what has already been described, the methods according to the embodiment and its variations described above may be used in suitable combinations.

[0117] In addition, although not illustrated individually, the present invention may be implemented with various modifications without departing from the scope of the invention.

Hardware Configurations of the Terminal Device and Printer

[0118] Next, sample hardware configurations for the terminal device **100** and printer **200** will be described with reference to FIG. 6.

Terminal Device

[0119] As shown in FIG. 6, the terminal device **100** has the processor **110**, the storage **115**, a display **140**, an operating interface **150**, and a communication interface **190**. The processor **110**, storage **115**, display **140**, operating interface **150**, and communication interface **190** are interconnected via a bus **105** so as to be capable of exchanging data with each other.

[0120] The storage **115** is provided with a volatile storage **120** and a nonvolatile storage **130**. The volatile storage **120** is DRAM, for example, and stores various programs and data used for implementing processes. The nonvolatile storage **130** is a hard disk drive or a solid state drive, for example, and has a program storage area **131** and a data storage area **132**.

[0121] The program storage area **131** stores the OS **101** various programs, including a program of the OS **101**, the printer application **170**, the printing-related programs, and the like based on the flowcharts in FIGS. 2 through 5 described above. The data storage area **132** stores data and the like required for executing the various programs.

[0122] The processor **110** is a device that performs data processing, such as a CPU. The processor **110** executes the various programs stored in the program storage area **131**. The processor **110** performs various processes including data communications with the printer **200** connected to a network NT. The processor **110** executing one of the programs stored in the program storage area **131** is an example of the processor, an example of the controller, and an example of a computer.

[0123] The display **140** is a liquid crystal display, for example, and can display various information. The operating interface **150** is a user interface, and includes a mouse and keyboard, for example, and accepts user operations. By operating the operating interface **150**, the user can input various instructions into the terminal device **100**.

[0124] The communication interface **190** connects to the network NT for communicating with other devices. The communication interface **190** may be a network interface, or a direct interface for directly connecting an external device, such as a USB interface. The communication interface **190** is an example of the communication interface.

[0125] The storage **115** is not limited to a configuration having the above components but may be configured of RAM, ROM, EEPROM, an HDD, a portable storage medium such as a USB memory device that is mounted in the terminal device **100**, a buffer provided in the processor **110**, and the like, or a combination of these components.

[0126] The storage **115** may also be a computer-readable storage medium. A computer-readable storage medium is a non-transitory medium. In addition to the above examples, non-transitory media include CD-ROM and DVD-ROM. A non-transitory medium is also a tangible medium. The same applies to a storage **215** of the printer **200** described later.

Printer

[0127] As shown in FIG. 6, the printer **200** has a processor **210**, a storage **215**, a display **240**, an operating interface **250**, a communication interface **270**, and a print engine **290**. The processor **210**, storage **215**, display **240**, operating interface **250**, communication interface **270**, and print engine **290** are interconnected via a bus **205** so as to be capable of exchanging data with each other.

[0128] The storage **215** is provided with a volatile storage **220** and a nonvolatile storage **230**. The volatile storage **220** is DRAM, for example. The volatile storage **220** is provided with an image data storage area **222** for storing image data to be printed. The nonvolatile storage **230** is NVRAM or flash memory, for example. The nonvolatile storage **230** is provided with a program storage area **231** and a data storage area **232**. The program storage area **231** stores various programs including the printing-related programs and the like in the form of firmware or the like. The data storage area

232 stores various data and the like for executing the various programs.

[0129] The processor **210** is a device that performs data processing, such as a CPU. The processor **210** executes the various programs stored in the program storage area **231**.

[0130] The display **240** is a liquid crystal display, for example, and can display various information. The operating interface **250** is a user interface that is a device that accepts user operations. By operating the operating interface **250**, the user can input various instructions into the printer **200**. The communication interface **270** is a wired or wireless network interface for communicating with other devices. The communication interface **270** is connected to the network NT. The communication interface **270** may be a direct interface for directly connecting an external device, such as a USB interface.

[0131] The print engine **290** prints images on sheets conveyed by a conveying mechanism (not shown). The print engine **290** prints images based on print data received from the terminal device **100** or print jobs generated by the printer **200** itself in accordance with operations received through the operating interface **250**. The print engine **290** may have a thermal head to print images using a thermal transfer system. However, the print engine **290** may employ a printing system other than the thermal transfer system such as such as an inkjet printing system and an electrophotographic system. The print engine **290** may be referred to as the printing member **290** or the printing assembly **290**.

Claims

1. A non-transitory computer readable storage medium storing an information processing program executed by a computer included in an information processing device, the information processing device having an operating system installed thereon, the operating system having standard printing software integrated therein, wherein the standard printing software is configured to perform a modification process when a size of an original image represented by original print data exceeds a designated sheet size of a printing sheet for use in printing the original print data, wherein the modification process includes modifying the original print data to size-reduced print data representing a size-reduced image derived by reducing the size of the original image, the information processing program supporting a printer communicable with the information processing device, the information processing program, when executed by the computer, causing the information processing device to perform: receiving specified print data from the standard printing software, the specified print data being one of the original print data and the size-reduced print data; in response to reception of the specified print data, transmitting, to the standard printing software, a request for the original print data; receiving the original print data from the standard printing software as a response to the request; and transmitting, to the printer, finalized print data based on the original print data received as the response to the request.
2. The non-transitory computer readable storage medium according to claim 1, wherein the transmitting the request is performed regardless of the designated sheet size.
3. The non-transitory computer readable storage medium according to claim 1, wherein the information processing program, when executed by the computer, causes the information processing device to further perform: acquiring the designated sheet size; and determining whether the size of the original image represented by the original print data exceeds the designated sheet size, wherein the transmitting the request is performed regardless of the designated sheet size, wherein the transmitting the finalized print data is performed when the determining determines that the size of the original image represented by the original print data exceeds the designated sheet size.
4. The non-transitory computer readable storage medium according to claim 3, wherein the information processing program, when executed by the computer, causes the information processing device to further perform: when the determining determines that the size of the original

image represented by the original print data is less than or equal to the designated sheet size, transmitting, to the printer, second finalized print data based on the specified print data, wherein the transmitting the finalized print data is not performed when the determining determines that the size of the original image represented by the original print data is less than or equal to the designated sheet size.

5. The non-transitory computer readable storage medium according to claim 1, wherein the information processing program, when executed by the computer, causes the information processing device to further perform: acquiring the designated sheet size; and determining whether the designated sheet size is associated with a specific sheet size having an indefinite length, wherein the transmitting the request is performed when the determining determines that the designated sheet size is associated with the specific sheet size.

6. The non-transitory computer readable storage medium according to claim 5, wherein the information processing program, when executed by the computer, causes the information processing device to further perform: when the determining determines that the designated sheet size is not associated with the specific sheet size, transmitting, to the printer, second finalized print data based on the specified print data, wherein the transmitting the finalized print data is not performed when the determining determines that the designated sheet size is not associated with the specific sheet size.

7. The non-transitory computer readable storage medium according to claim 5, wherein the transmitting the request is not performed when the determining determines that the designated sheet size is not associated with the specific sheet size.

8. The non-transitory computer readable storage medium according to claim 1, wherein the information processing program is registered as a virtual printer in the standard printing software so that the standard printing software can transmit the original print data and the size-reduced print data to the information processing program.

9. The non-transitory computer readable storage medium according to claim 1, wherein the finalized print data is for use in printing an image on the printing sheet of a continuous length sheet wound in a roll to create a label.

10. The non-transitory computer readable storage medium according to claim 1, wherein the finalized print data represents a finalized image having the same size as the original image.

11. The non-transitory computer readable storage medium according to claim 1, wherein the information processing program, when executed by the computer, causes the information processing device to further perform: acquiring an indicator indicating whether a user selected sheet size includes an indefinite dimension; and determining whether the indicator indicates that the user selected sheet size includes the indefinite dimension, wherein the transmitting the request is performed when the determining determines that the indicator indicates that the user selected sheet size includes the indefinite dimension.

12. An information processing device comprising: a communication interface communicable with a printer; a storage; and a processor; wherein an information processing program supporting the printer and stored in the storage, when executed by the processor, causes the processor to perform: receiving specified print data from a standard printing software, the standard printing software integrated in an operating system installed on the information processing device, wherein the standard printing software is configured to perform a modification process when a size of an original image represented by original print data exceeds a designated sheet size of a printing sheet for use in printing the original print data, wherein the modification process includes modifying the original print data to size-reduced print data representing a size-reduced image derived by reducing the size of the original image, the specified print data being one of the original print data and the size-reduced print data; in response to reception of the specified print data, transmitting, to the standard printing software, a request for the original print data; receiving the original print data from the standard printing software as a response to the request; and transmitting, to the printer

through the communication interface, finalized print data based on the original print data received as the response to the request.
